TENSORING A DG MODULE WITH A COMPLEX OF MODULES GIVES A FUNCTOR TO COMPLEXES OF SHEAVES

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Editorial note. The counted statement is the author's own lemma environment labelled lemma-tensor-with-complex (source lines 6118-6129); the proof environment begins two lines below it and is its own proof, closing at source line 6153.

Editorial note. The proof invokes the author's lemma lemma-tensor by label; that lemma's statement and its own proof are retained as a context span placed before the counted statement, and the generator expands the context transitively, so the reference resolves inside the assembled package and no source pointer is declared.

Editorial note. The statement and the proof both display the tensor functor notation; the displays are part of the retained source range and are reproduced as written, so no figure file is needed.

Editorial note. The retained preamble is the chapter's own minimal preamble (sources/stacks-dga/preamble.tex, taken with its real line count), a verbatim excerpt of the native Xy-pic setup, the native theorem declarations with the mathematical macros, and the project's own no-op reference environments; no macro was invented or redefined.

Editorial scope. One original result of the Stacks Project's Differential Graded Algebra chapter, counted once: tensoring with a complex of O-modules is a functor from dg E-modules to complexes of O-modules, sending E to the complex and preserving direct sums.

Allowed standard prerequisites (editorial).

- Dg E-module and dg bimodule. A module over the dg algebra E together with a differential, and a complex carrying compatible left E-module and right O(U)-module structures.. The proof observes that each K(U) is a dg (E, O(U))-bimodule (source lines 6132-6135).
- Tensor functor with a dg bimodule. Tensoring with a dg bimodule is a functor of dg categories, sending the algebra to the bimodule and preserving direct sums (the author's lemma lemma-tensor, retained as context).. It supplies the pointwise functors and the preservation of direct sums (source lines 6136-6138, 6149-6151).
- Complexes of O-modules on a site. The dg category of complexes of O-modules on the site under consideration, with the pointwise evaluation functors.. It is the target category of the constructed functor (source lines 6122-6126, 6145-6148).

Date: stacks-project:dga.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *Let R be a ring. Let (A, d) and (B, d) be differential graded algebras over R. Let N be a differential graded (A, B)-bimodule. Then $M \mapsto M \otimes_A N$ defines a functor*

$$- \otimes_A N : Mod^{dg}_{(A,d)} \longrightarrow Mod^{dg}_{(B,d)}$$

of differential graded categories. This functor induces functors

$$Mod_{(A,d)} \to Mod_{(B,d)} \quad and \quad K(Mod_{(A,d)}) \to K(Mod_{(B,d)})$$

by an application of Lemma 2.1.

Proof. Above we have seen how the construction defines a functor of underlying graded categories. Thus it suffices to show that the construction is compatible with differentials. Let M and M' be differential graded A-modules and let $f : M \to M'$ be an A-module homomorphism which is homogeneous of degree n. Then we have

$$\mathrm{d}(f) = \mathrm{d}_{M'} \circ f - (-1)^n f \circ \mathrm{d}_M$$

On the other hand, we have

$$\mathrm{d}(f \otimes \mathrm{id}_N) = \mathrm{d}_{M' \otimes_A N} \circ (f \otimes \mathrm{id}_N) - (-1)^n (f \otimes \mathrm{id}_N) \circ \mathrm{d}_{M \otimes_A N}$$

Applying this to an element $x \otimes y$ with $x \in M$ and $y \in N$ homogeneous we get

$$\begin{aligned} \mathrm{d}(f \otimes \mathrm{id}_N)(x \otimes y) =& \mathrm{d}_{M'}(f(x)) \otimes y + (-1)^{n+\deg(x)} f(x) \otimes \mathrm{d}_N(y) \\ & - (-1)^n f(\mathrm{d}_M(x)) \otimes y - (-1)^{n+\deg(x)} f(x) \otimes \mathrm{d}_N(y) \\ =& \mathrm{d}(f)(x \otimes y) \end{aligned}$$

Thus we see that $\mathrm{d}(f) \otimes \mathrm{id}_N = \mathrm{d}(f \otimes \mathrm{id}_N)$ and the proof is complete. □

1. Bimodules

2. Tensor product

Lemma 2.1. *Let R be a ring. A functor $F : \mathcal{A} \to \mathcal{B}$ of differential graded categories over R induces functors $Comp(\mathcal{A}) \to Comp(\mathcal{B})$ and $K(\mathcal{A}) \to K(\mathcal{B})$.*

Proof. Omitted. □

Lemma 2.2. *In the situation above there is a functor*

$$- \otimes_E K^\bullet : Mod^{dg}_{(E,d)} \longrightarrow Comp^{dg}(\mathcal{O})$$

of differential graded categories. This functor sends E to $K^\bullet$ and commutes with direct sums.

Proof. Let M be a differential graded E-module. For every object U of $\mathcal{C}$ the complex $K^\bullet(U)$ is a left differential graded E-module as well as a right $\mathcal{O}(U)$-module. The actions commute, so we have a bimodule. Thus, by the constructions in Sections 2 and 1 we can form the tensor product

$$M \otimes_E K^\bullet(U)$$

which is a differential graded $\mathcal{O}(U)$-module, i.e., a complex of $\mathcal{O}(U)$-modules. This construction is functorial with respect to U, hence we can sheafify to get a complex of $\mathcal{O}$-modules which we denote

$$M \otimes_E K^\bullet$$

Moreover, for each U the construction determines a functor $\mathrm{Mod}^{dg}_{(E,\mathrm{d})} \to \mathrm{Comp}^{dg}(\mathcal{O}(U))$ of differential graded categories by Lemma 0.1. It is therefore clear that we obtain a functor as stated in the lemma. □